

Chapter 8

Rigid Body Motion

Introduction

- An airplane propeller, a revolving door, a ceiling fan, and a Ferris wheel all involve rotating rigid objects.
- Real-world rotations can be very complicated because of stretching and twisting of the rotating body. But for now we'll assume that the rotating body is perfectly rigid.



What is a rigid body?

There are cases where an object **cannot** be treated as a particle. In these cases the **size** or **shape** of the body must be considered.

Rotation of the body about its center of mass requires a different approach.

For example, in the design of gears, cams, and links in machinery or mechanisms, rotation of the body is an important aspect in the analysis of motion.

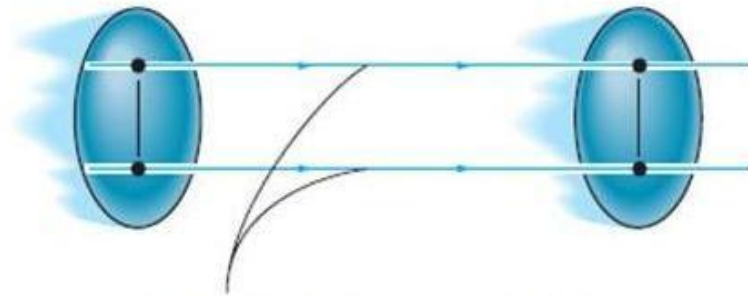
Rigid body motion

We will now start to study **rigid body motion**. The analysis will be limited to **planar motion**.

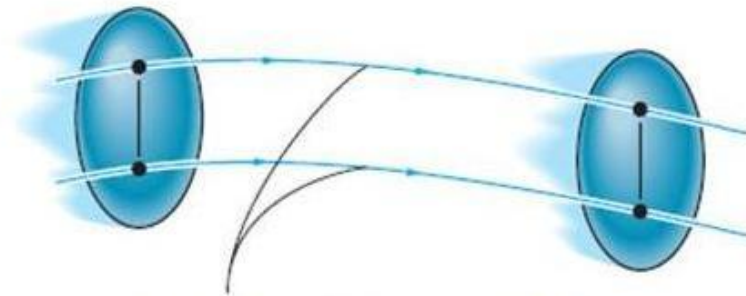
A body is said to undergo planar motion when all parts of the body move along paths equidistant from a fixed plane.

Planar rigid body motion

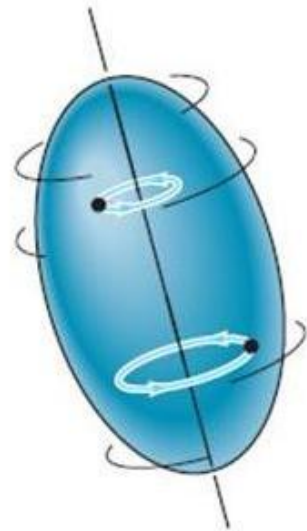
There are **three** types of planar rigid body motion.



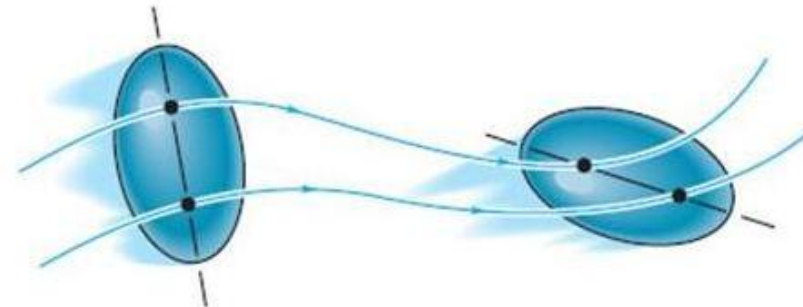
Path of rectilinear translation



Path of curvilinear translation

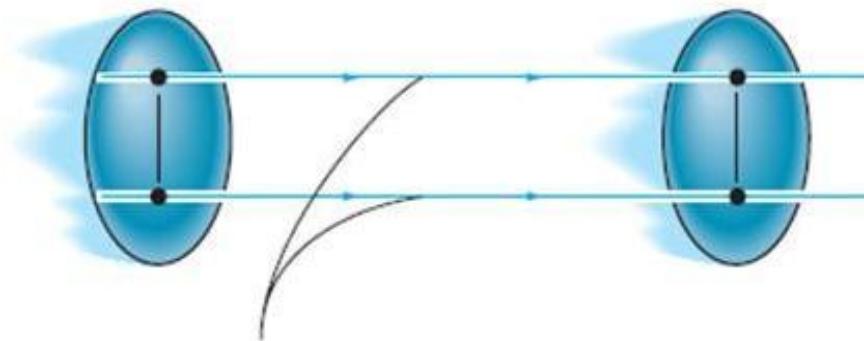


Rotation about a fixed axis

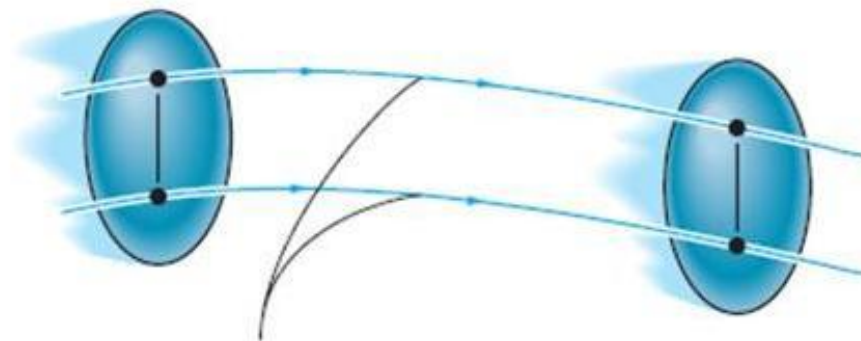


General plane motion

Planar rigid body motion



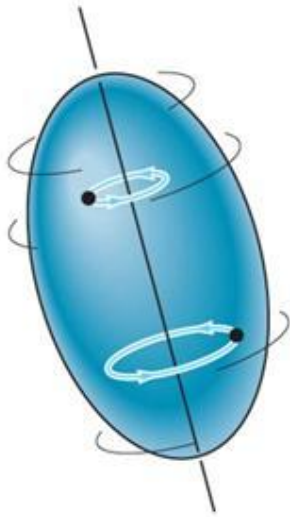
Path of rectilinear translation



Path of curvilinear translation

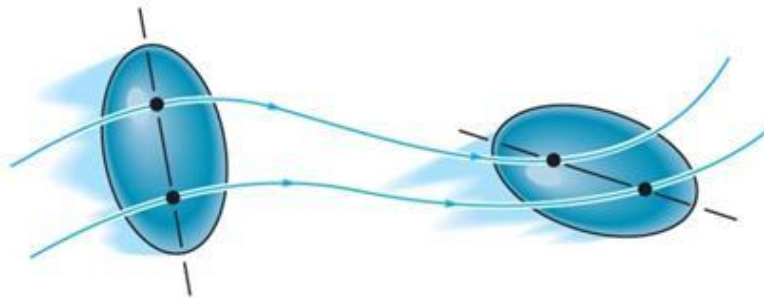
Translation: Translation occurs if every line segment on the body remains parallel to its original direction during the motion. When all points move along straight lines, the motion is called **rectilinear** translation. When the paths of motion are curved lines, the motion is called **curvilinear** translation.

Planar rigid body motion



Rotation about a fixed axis

Rotation about a fixed axis: In this case, all the particles of the body, except those on the axis of rotation, move along **circular paths** in planes perpendicular to the axis of rotation.

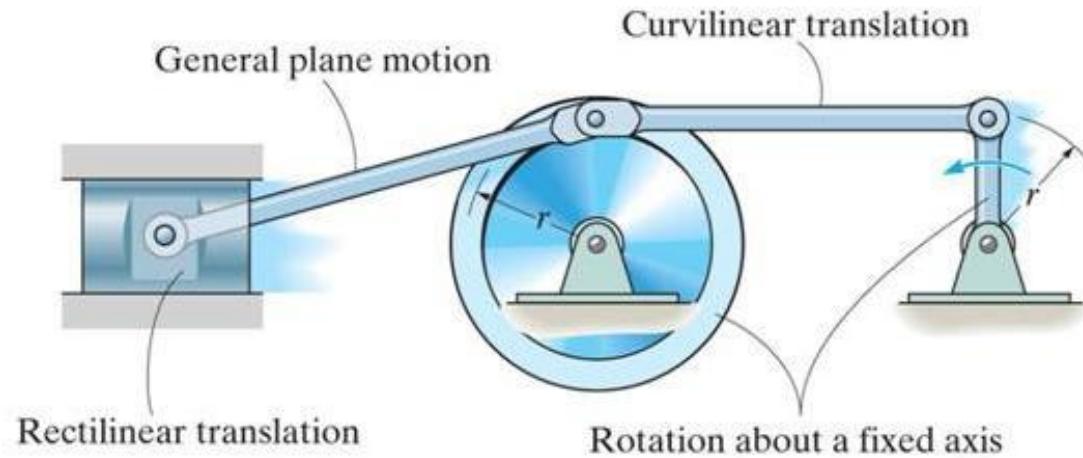


General plane motion

General plane motion: In this case, the body undergoes **both translation and rotation**. Translation occurs within a plane and rotation occurs about an axis ***perpendicular*** to this plane.

Planar rigid body motion

An example of bodies undergoing the three types of motion is shown in this mechanism.



The wheel and crank undergo **rotation about a fixed axis**. In this case, both axes of rotation are at the location of the pins and perpendicular to the plane of the figure.

The piston undergoes **rectilinear translation** since it is constrained to slide in a straight line.

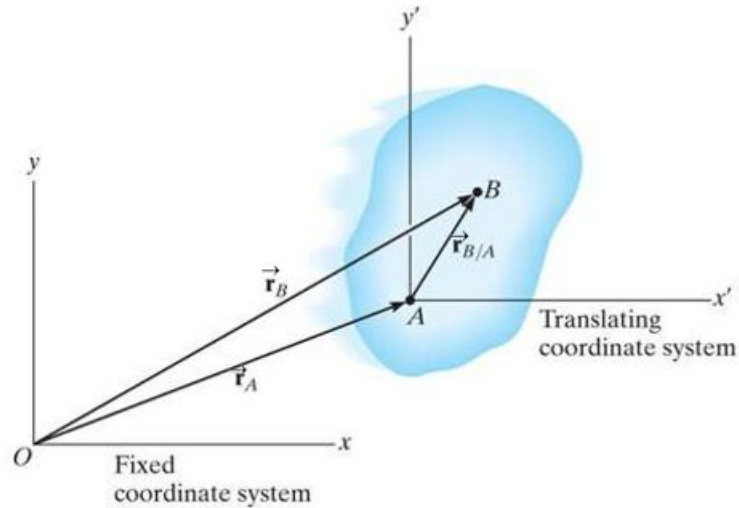
The connecting rod on the right undergoes **curvilinear translation**, since it will remain horizontal as it moves along a circular path. The connecting rod on the left undergoes **general plane motion**, as it will both translate and rotate.

The velocity at B is $\mathbf{v}_B = \mathbf{v}_A + d\mathbf{r}_{B/A}/dt$

Now $d\mathbf{r}_{B/A}/dt = 0$ since $\mathbf{r}_{B/A}$ is constant. So, $\mathbf{v}_B = \mathbf{v}_A$, and by following similar logic, $\mathbf{a}_B = \mathbf{a}_A$.
The positions of two points A and B on a translating body can be related by

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$$

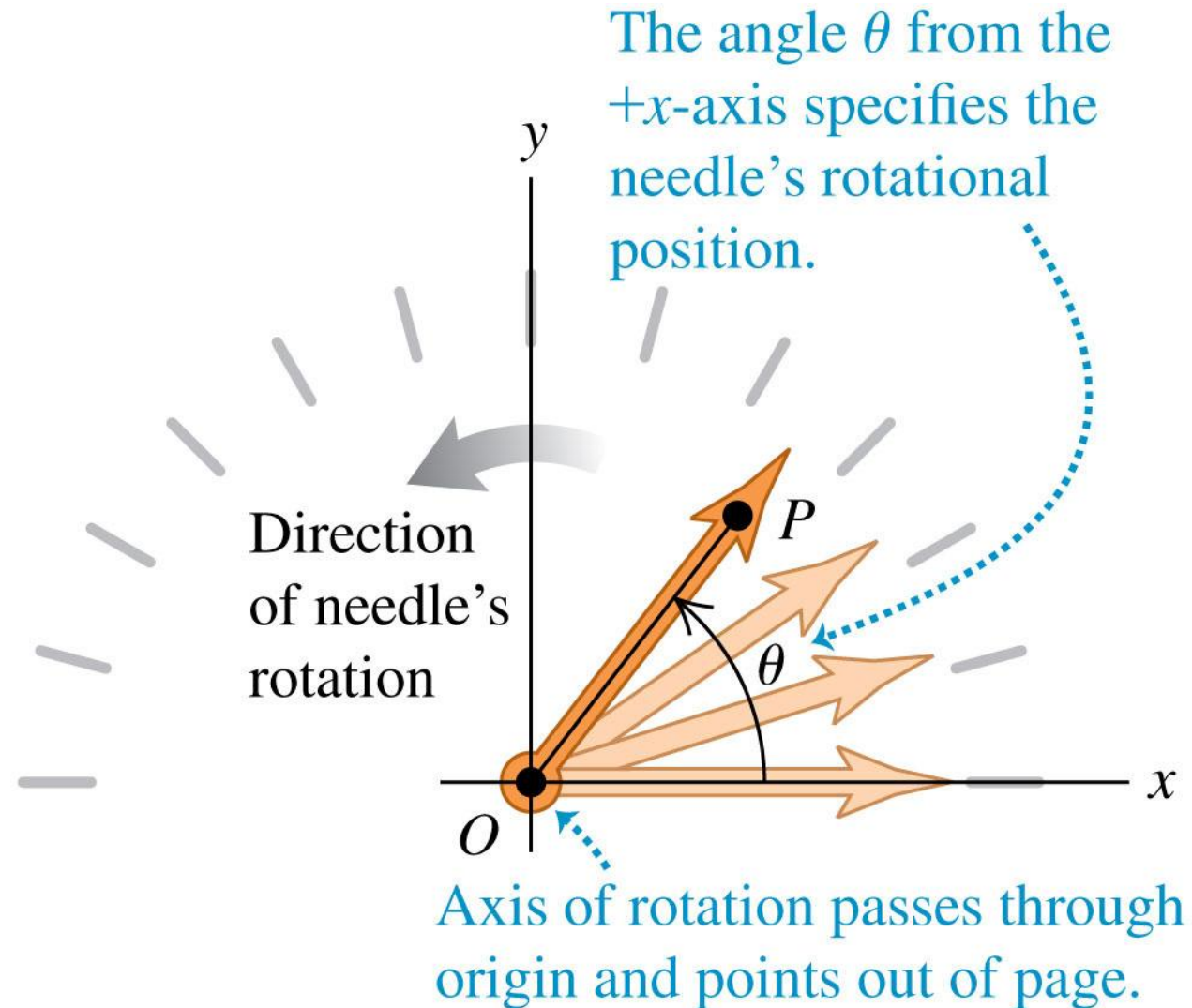
where $\mathbf{r}_B$ & $\mathbf{r}_A$ are the absolute position vectors defined from the fixed x-y coordinate system, and $\mathbf{r}_{B/A}$ is the relative-position vector between B and A.



Note, all points in a rigid body subjected to translation move with the same velocity and acceleration.

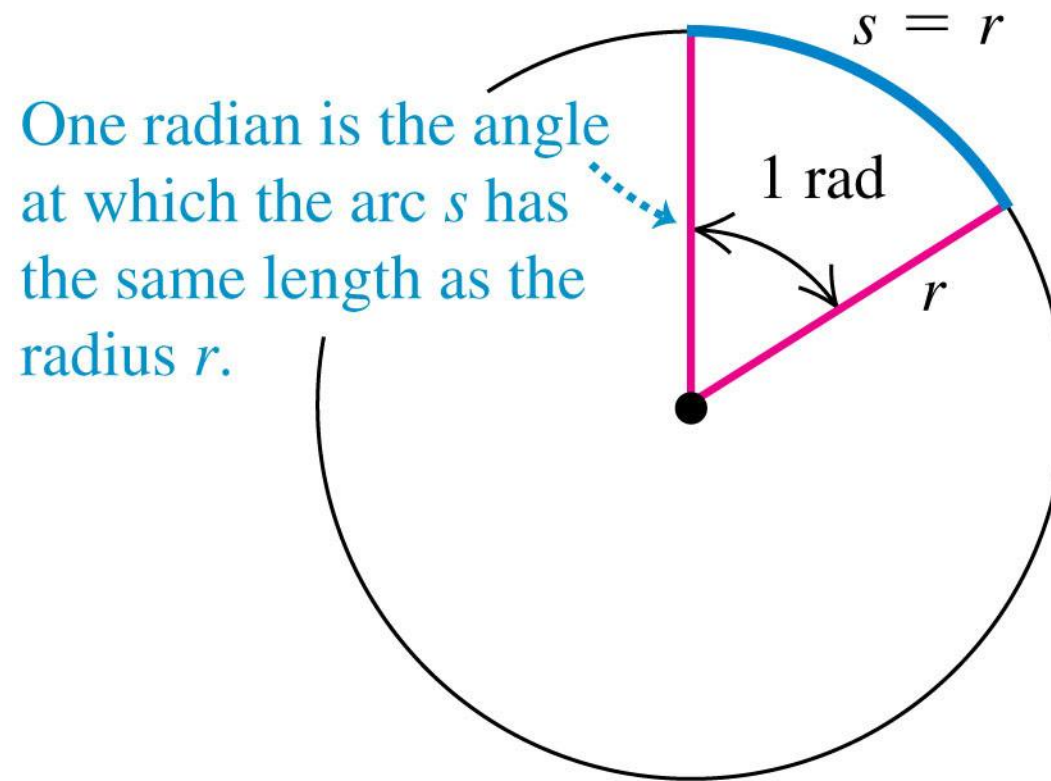
Angular coordinate

- A car's speedometer needle rotates about a *fixed axis*.



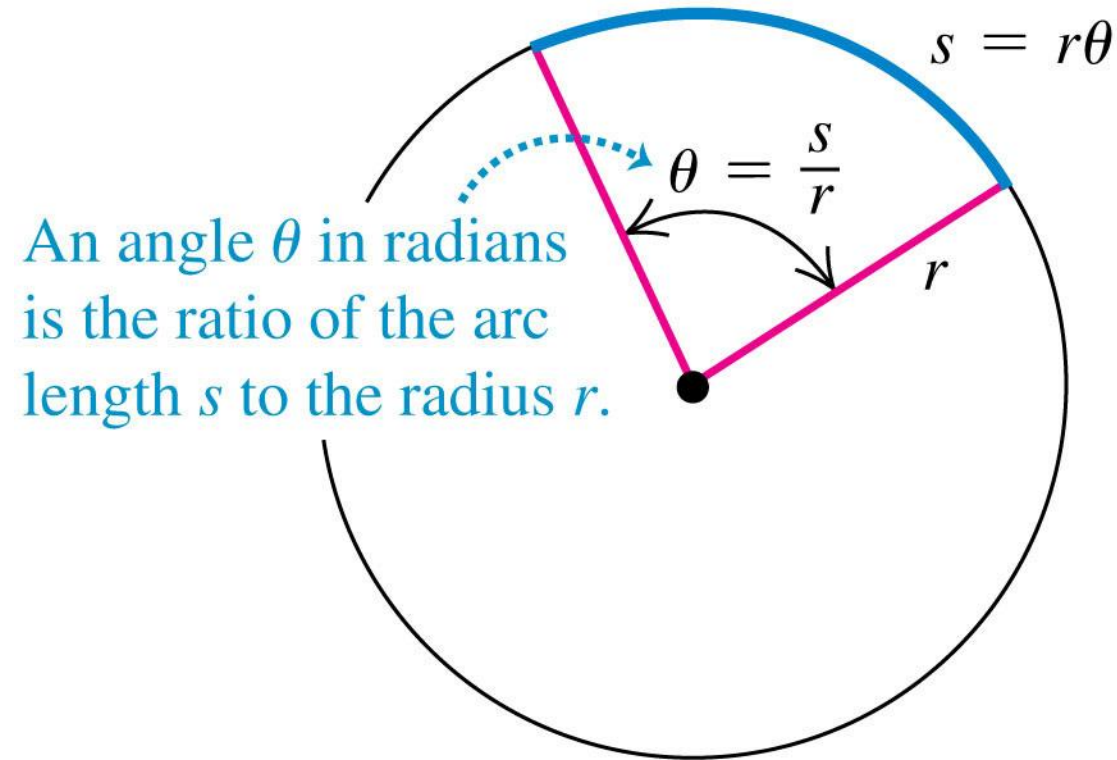
Units of angles

- One complete revolution is $360^\circ = 2\pi$ radians.



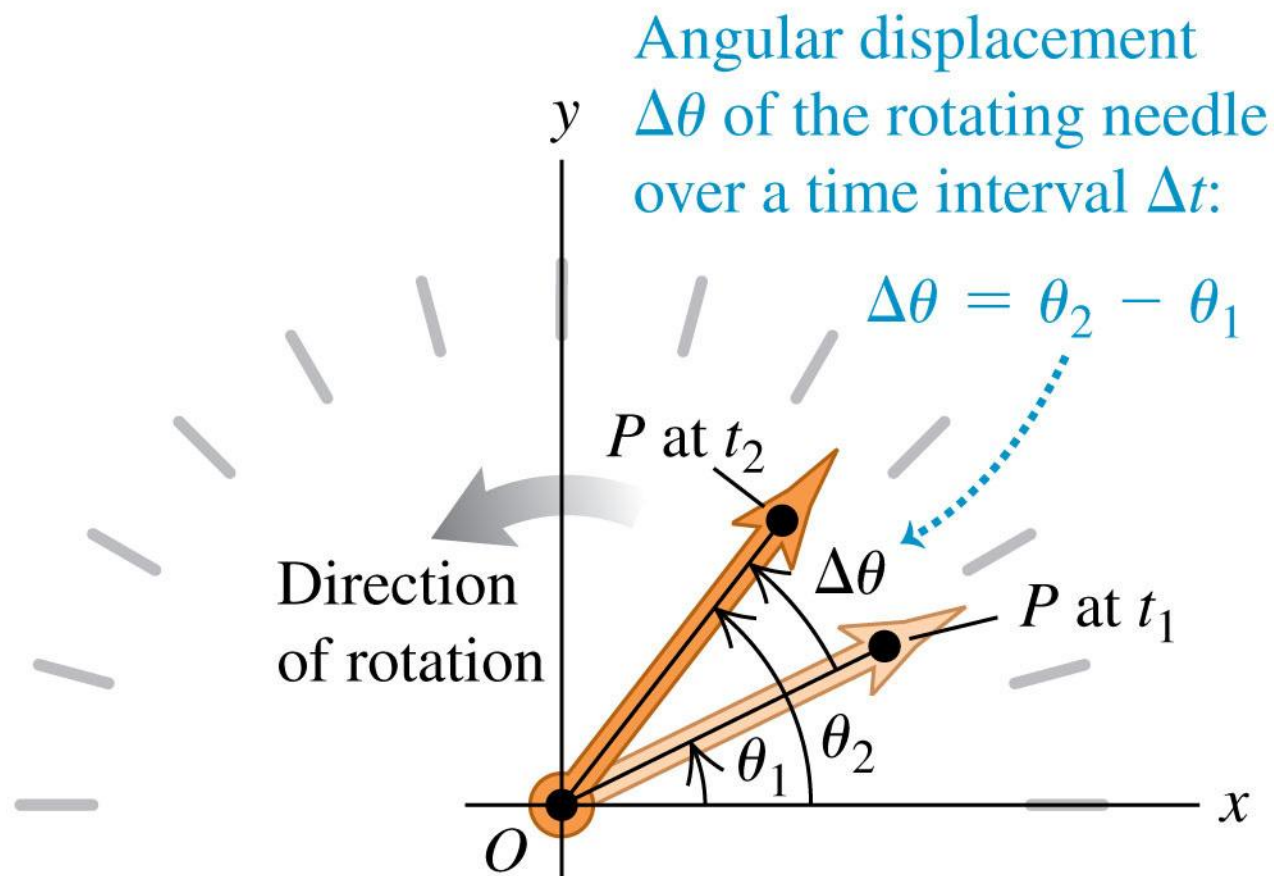
Units of angles

- An angle in radians is $\theta = s/r$, as shown in the figure.



Angular velocity

- The *average angular velocity* of a body is $\omega_{\text{av-z}} = \Delta\theta/\Delta t$.
- The subscript z means that the rotation is about the z -axis.



Angular velocity

- We choose the angle θ to increase in the counterclockwise rotation.

Counterclockwise rotation:

θ increases, so angular velocity is positive.

$\Delta\theta > 0$, so

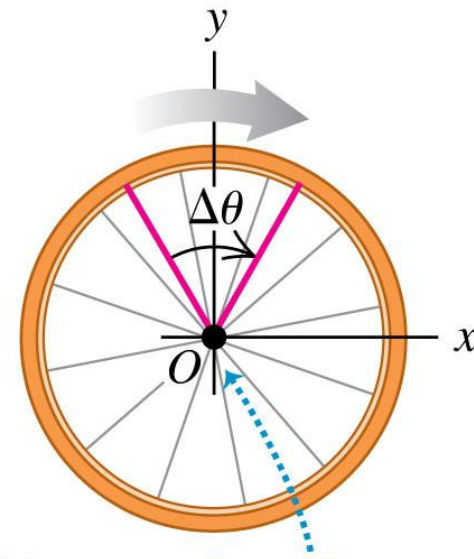
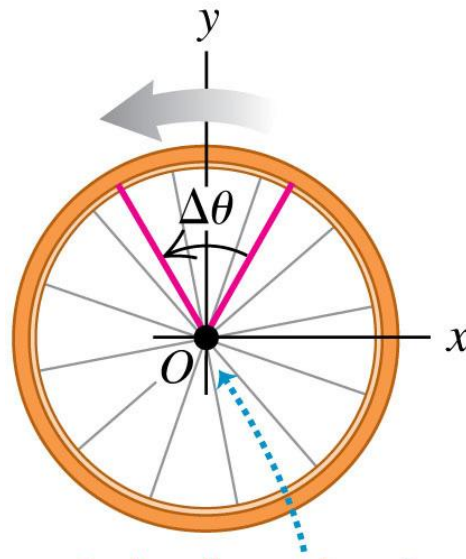
$$\omega_{\text{av-z}} = \Delta\theta/\Delta t > 0$$

Clockwise rotation:

θ decreases, so angular velocity is negative.

$\Delta\theta < 0$, so

$$\omega_{\text{av-z}} = \Delta\theta/\Delta t < 0$$



Axis of rotation (z-axis) passes through origin and points out of page.

Instantaneous angular velocity

- The **instantaneous angular velocity** is the limit of average angular velocity as $\Delta\theta$ approaches zero:

The **instantaneous angular velocity** of a rigid body rotating around the z -axis ...

$$\omega_z = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

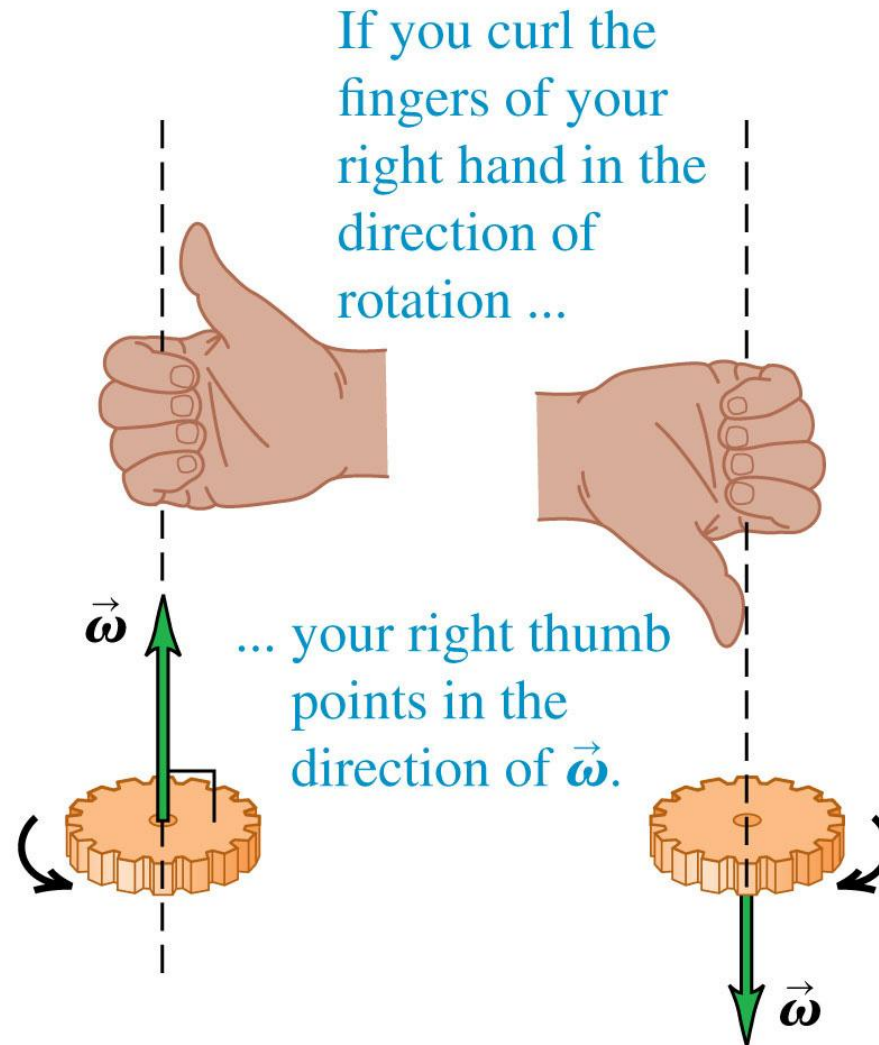
... equals the limit of the body's average angular velocity as the time interval approaches zero ...

... and equals the instantaneous rate of change of the body's angular coordinate.

- When we refer simply to “angular velocity,” we mean the instantaneous angular velocity, not the average angular velocity.
- The z -subscript means the object is rotating around the z -axis.
- The angular velocity can be positive or negative, depending on the direction in which the rigid body is rotating.

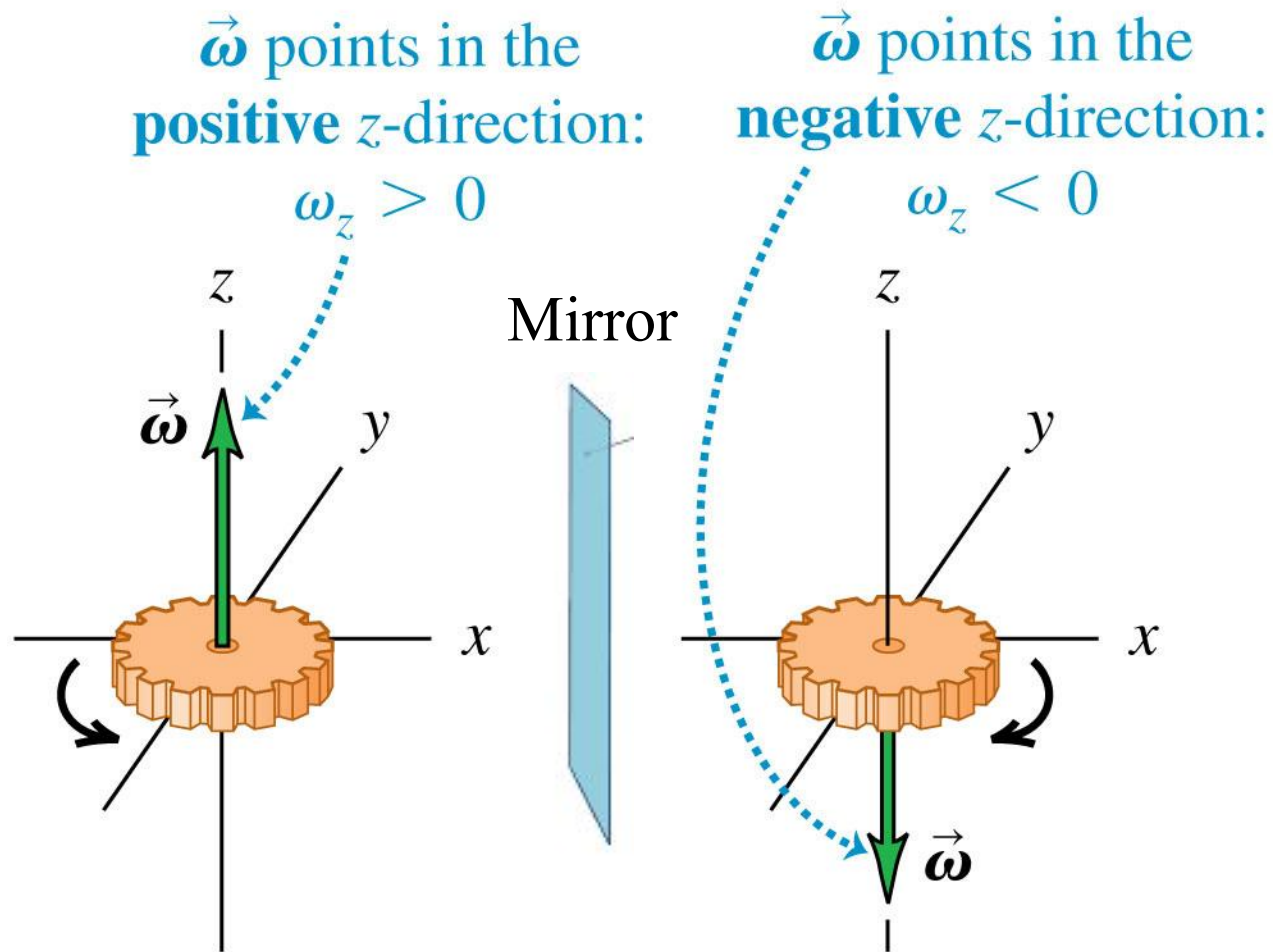
Angular velocity is a vector

- Angular velocity is defined as a vector whose direction is given by the right-hand rule.



Angular velocity is a pseudovector

- The sign of ω_z for rotation along the z -axis.
- The direction is reversed under parity.

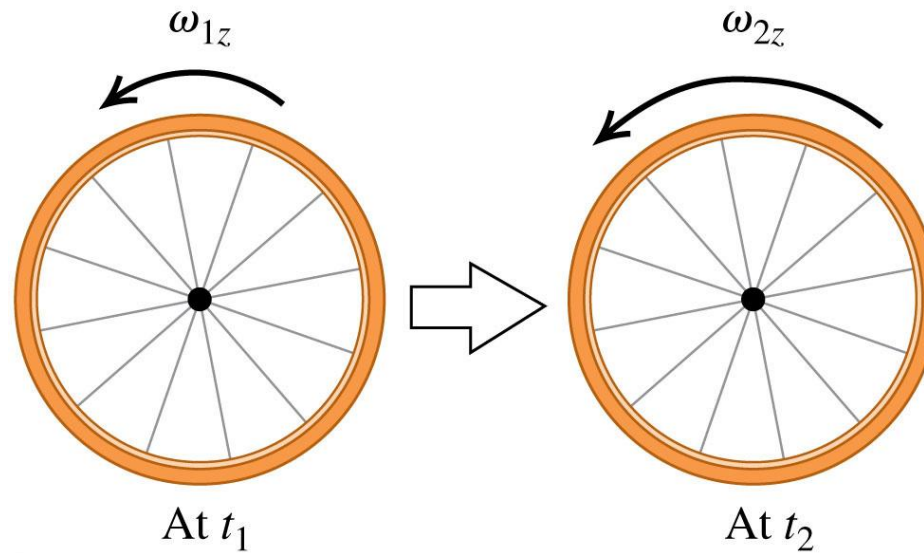


Angular acceleration

The instantaneous angular acceleration is $\alpha_z = d\omega_z/dt$.

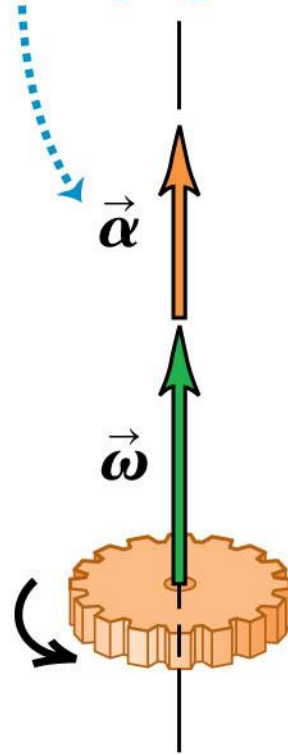
The average angular acceleration is the change in angular velocity divided by the time interval:

$$\alpha_{\text{av-}z} = \frac{\omega_{2z} - \omega_{1z}}{t_2 - t_1} = \frac{\Delta\omega_z}{\Delta t}$$

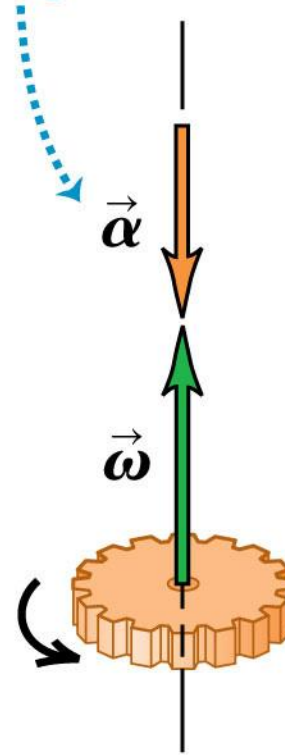


Angular acceleration as a vector

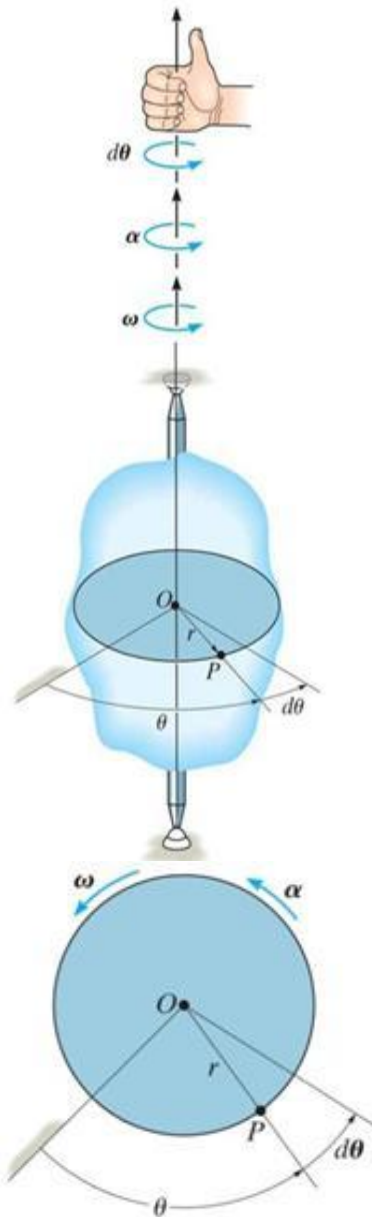
$\vec{\alpha}$ and $\vec{\omega}$ in the **same** direction: Rotation speeding up.



$\vec{\alpha}$ and $\vec{\omega}$ in the **opposite** directions: Rotation slowing down.



Rotation about a fixed axis



When a body rotates about a fixed axis, any point P in the body travels along a **circular path**. The angular position of P is defined by θ .

The change in angular position, $d\theta$, is called the angular displacement, with units of either radians or revolutions. They are related by
1 revolution = (2π) radians

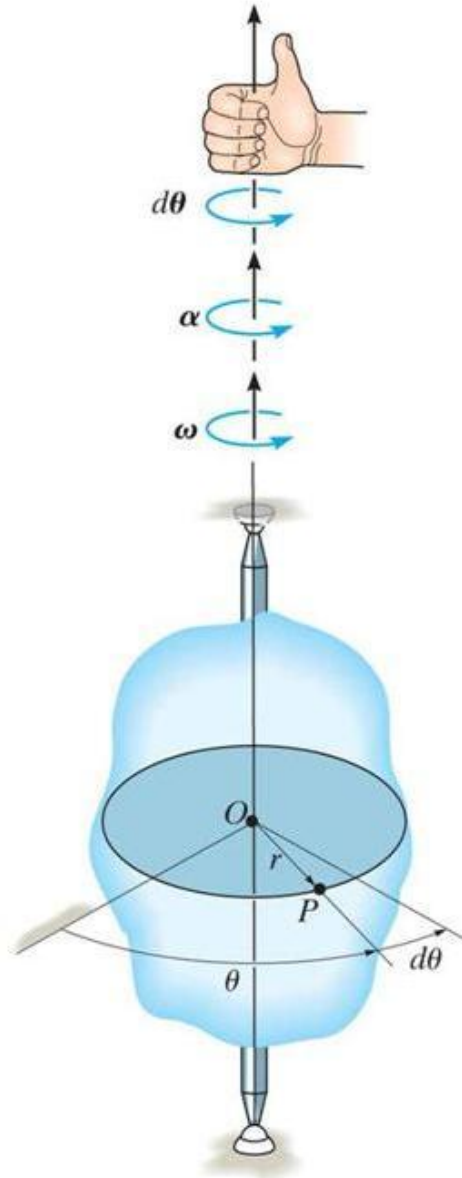
Angular velocity, ω , is obtained by taking the time derivative of angular displacement:

$$\omega = d\theta/dt$$

Similarly, **angular acceleration** is

$$\alpha = d^2\theta/dt^2 = d\omega/dt \text{ or } \alpha = \omega(d\omega/d\theta)$$

Rotation about a fixed axis



If the angular acceleration of the body is **constant**, $\alpha = \alpha_C$, the equations for angular velocity and acceleration can be integrated to yield the set of **algebraic** equations below.

$$\omega = \omega_0 + \alpha_C t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha_C t^2$$

$$\omega^2 = (\omega_0)^2 + 2\alpha_C (\theta - \theta_0)$$

θ_0 and ω_0 are the initial values of the body's angular position and angular velocity. Note these equations are very similar to the constant acceleration relations developed for the **rectilinear** motion of a particle.

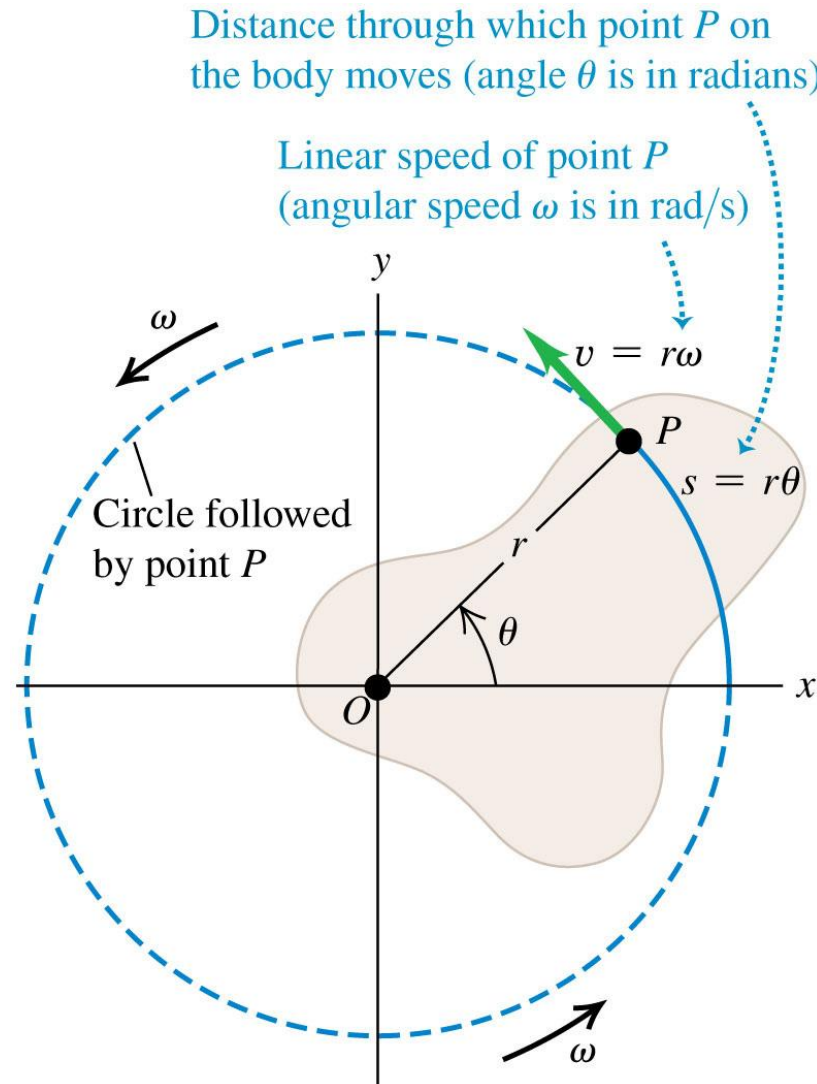
Rotation with constant angular acceleration

- The rotational formulas have the same form as the straight-line formulas, as shown in Table below.

Straight-Line Motion with Constant Linear Acceleration	Fixed-Axis Rotation with Constant Angular Acceleration
$a_x = \text{constant}$	$\alpha_z = \text{constant}$
$v_x = v_{0x} + a_x t$	$\omega_z = \omega_{0z} + \alpha_z t$
$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$	$\theta = \theta_0 + \omega_{0z} t + \frac{1}{2} \alpha_z t^2$
$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0)$
$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$	$\theta - \theta_0 = \frac{1}{2}(\omega_{0z} + \omega_z)t$

Relating linear and angular kinematics

- A point at a distance r from the axis of rotation has a linear speed of $v = r\omega$.



Relating linear and angular kinematics

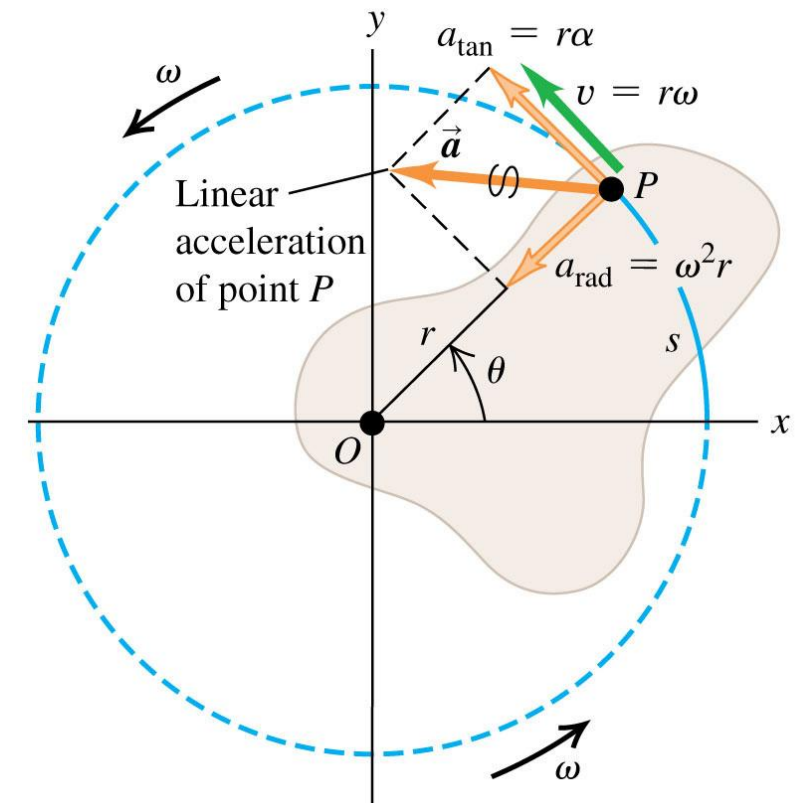
- For a point at a distance r from the axis of rotation:

- its tangential acceleration is $a_{\text{tan}} = r\alpha$;
- its centripetal (radial) acceleration is

$$a_{\text{rad}} = v^2/r = r\omega^2$$

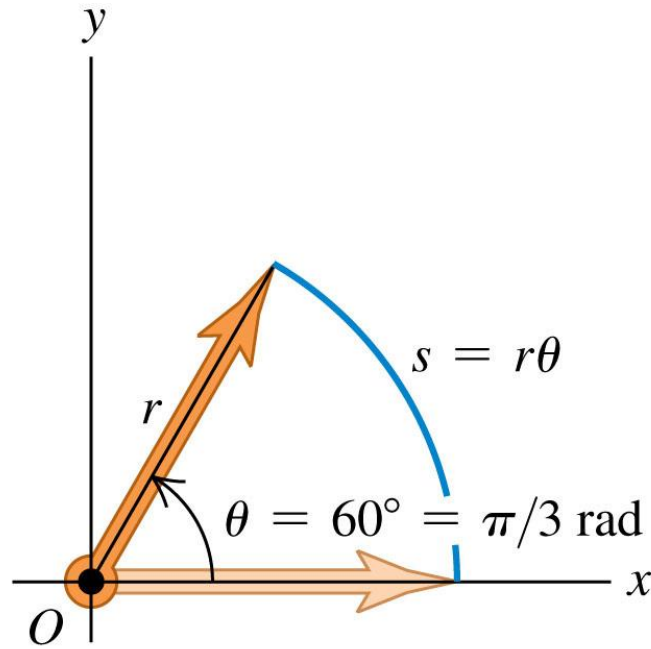
Radial and tangential acceleration components:

- $a_{\text{rad}} = \omega^2 r$ is point P 's centripetal acceleration.
- $a_{\text{tan}} = r\alpha$ means that P 's rotation is speeding up (the body has angular acceleration).



The importance of using radians, not degrees!

- Always use radians when relating linear and angular quantities.



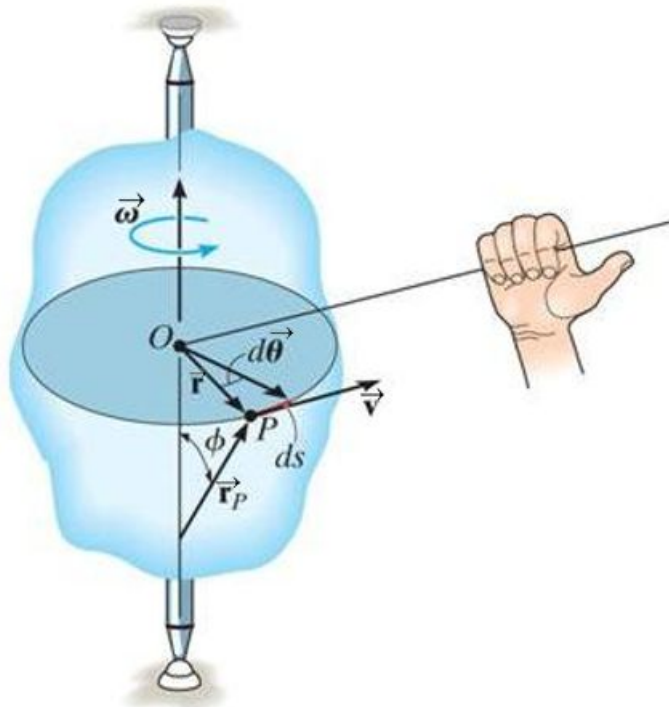
In any equation that relates linear quantities to angular quantities, the angles **MUST** be expressed in radians ...

RIGHT! $\blacktriangleright s = (\pi/3)r$

... never in degrees or revolutions.

WRONG $\blacktriangleright s = \cancel{60}r$

Rotation: velocity of point P



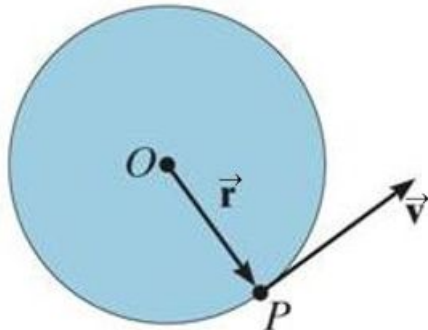
The magnitude of the velocity of P is equal to ωr . The velocity's direction is tangent to the circular path of P.

In the vector formulation, the magnitude and direction of $\mathbf{v}$ can be determined from the cross product of $\boldsymbol{\omega}$ and $\mathbf{r}_p$. Here $\mathbf{r}_p$ is a vector from any point on the axis of rotation to P.

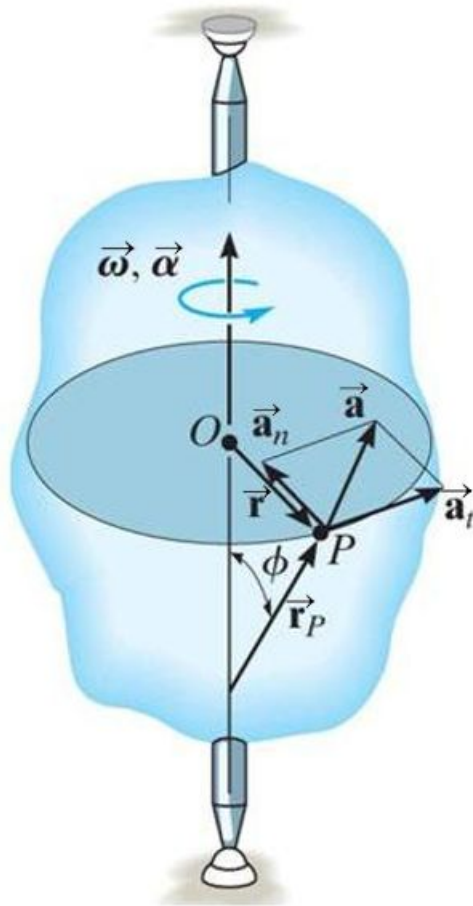
$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}_p = \boldsymbol{\omega} \times \mathbf{r} \quad \mathbf{r} \perp \mathbf{v}$$

The direction of $\mathbf{v}$ is determined by the right-hand rule.

$$\vec{\omega} = \frac{\vec{r} \times \vec{v}}{r^2}$$



Rotation: acceleration of point P



The acceleration of P is expressed in terms of its **normal** ($\mathbf{a}_n$) and **tangential** ($\mathbf{a}_t$) components.

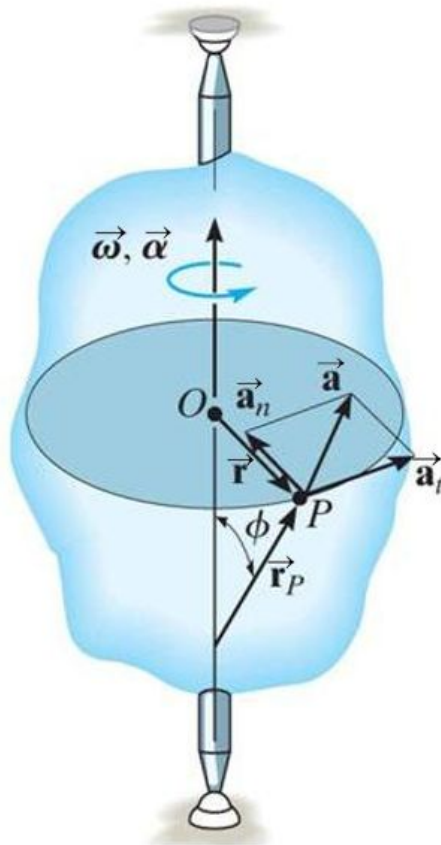
In scalar form, these are $a_t = r\alpha$ and $a_n = r\omega^2$.

The **tangential component**, $\mathbf{a}_t$, represents the time rate of change in the velocity's **magnitude**. It is directed **tangent** to the path of motion.

The **normal component**, $\mathbf{a}_n$, represents the time rate of change in the velocity's **direction**. It is directed **towards** the **center** of the circular path.

It can be shown that this equation reduces to

Rotation: acceleration of point P



Using the **vector** formulation, the acceleration of P can also be defined by differentiating the velocity.

$$\vec{a} = d(\vec{\omega} \times \vec{r}_p)/dt = (d\vec{\omega}/dt) \times \vec{r}_p + \vec{\omega} \times (d\vec{r}_p/dt) = \vec{\alpha} \times \vec{r}_p + \vec{\omega} \times (\vec{\omega} \times \vec{r}_p)$$

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$$

$$\vec{a} = \vec{\alpha} \times \vec{r}_p - \omega^2 \vec{r}_p \equiv \vec{a}_t + \vec{a}_n$$

The **magnitude** of the acceleration vector is $a = \sqrt{(a_t)^2 + (a_n)^2}$

Derivation

Rotation about a fixed axis: procedure

- Establish a **sign convention** along the axis of rotation.
- If a relationship is known between any **two** of the variables (α , ω , θ , or t), the other variables can be determined from the equations:
 $\omega = d\theta/dt \quad \alpha = d\omega/dt \quad \alpha d\theta = \omega d\omega$
- If α is **constant**, use the equations for constant angular acceleration.
- To determine the motion of a point, the scalar equations

$$v = r\omega, \quad a_t = r\alpha, \quad a_n = r\omega^2, \quad a = (a_t^2 + a_n^2)^{1/2}$$

- Or the vector equation

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}_p = \boldsymbol{\omega} \times \mathbf{r}$$

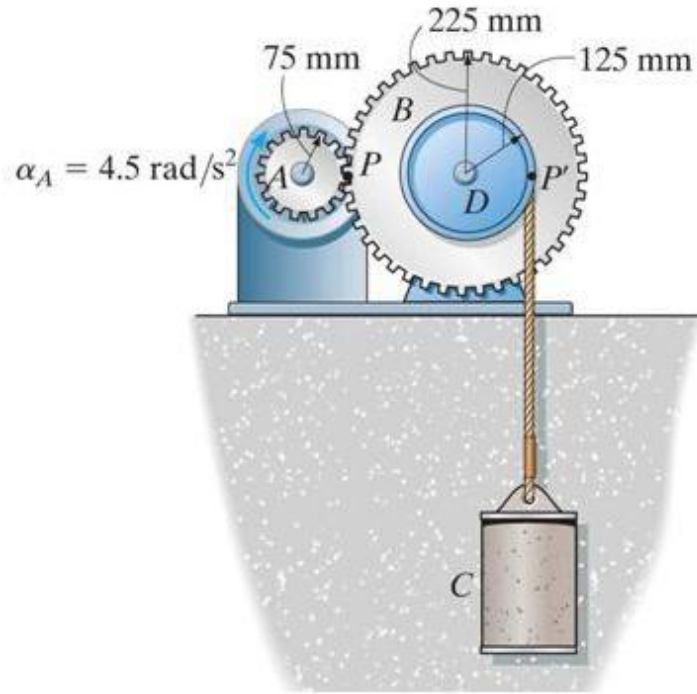
$$\mathbf{a} = \mathbf{a}_t + \mathbf{a}_n = \boldsymbol{\alpha} \times \mathbf{r} - \omega^2 \mathbf{r}$$

Quiz

1. If a rigid body is in translation only, the velocity at points A and B on the rigid body _____.
 - A) are usually different
 - B) are always the same
 - C) depend on their position
 - D) depend on their relative position

2. If a rigid body is rotating with a constant angular velocity about a fixed axis, the velocity vector at point P is _____.
 - A) $\vec{\omega} \times \vec{r}_p$
 - B) $\vec{r}_p \times \vec{\omega}$
 - C) $d\vec{r}_p/dt$
 - D) All of the above.

Example



Given: The motor turns gear A with a constant angular acceleration, $\alpha_A = 4.5 \text{ rad/s}^2$, starting from rest. The cord is wrapped around pulley D which is rigidly attached to gear B.

Find: The velocity of cylinder C and the distance it travels in 3 seconds.

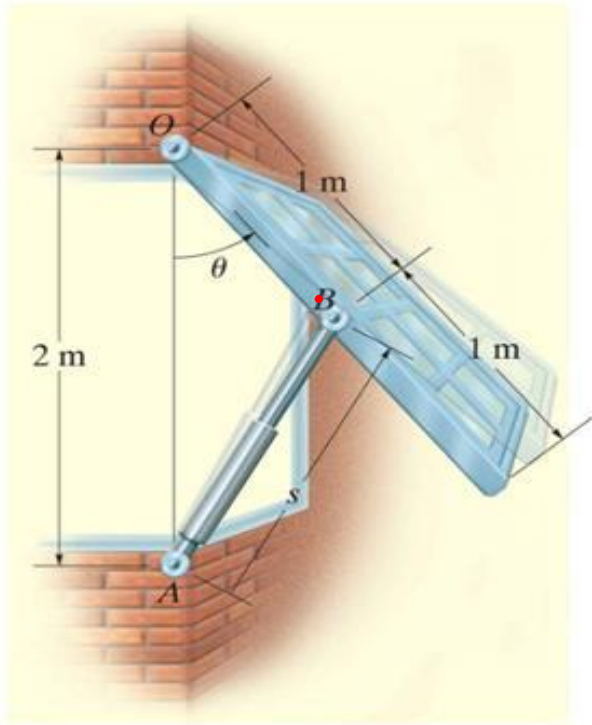
Quiz

1. A body subjected to general plane motion undergoes a/an
 - A) translation.
 - B) rotation.
 - C) simultaneous translation and rotation.
 - D) out-of-plane movement.

2. In general plane motion, if the rigid body is represented by a slab, the slab rotates
 - A) about an axis perpendicular to the plane.
 - B) about an axis parallel to the plane.
 - C) about an axis lying in the plane.
 - D) None of the above.

Absolute motion analysis

The figure below shows the window using a hydraulic cylinder AB.



The **absolute motion analysis method** relates the position of a point, B, on a rigid body undergoing rectilinear motion to the angular position, θ , of a line contained in the body.

Once a relationship in the form of $s_B = f(\theta)$ is established, the velocity and acceleration of point B are obtained in terms of the angular velocity and angular acceleration of the rigid body by taking the **first and second time derivatives** of the position function.

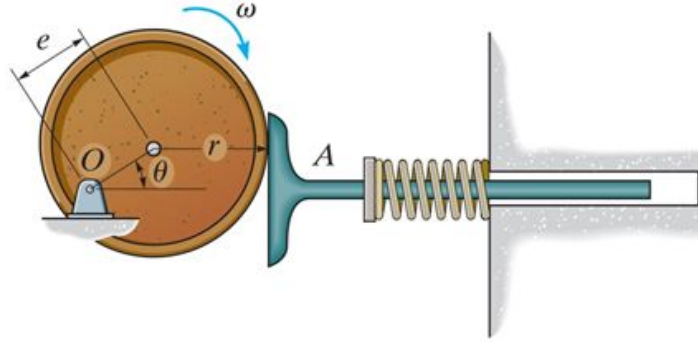
Usually the **chain rule** must be used when taking the derivatives of the position coordinate equation.

Procedure for absolute motion analysis

The velocity and acceleration of a point undergoing rectilinear motion can be related to the angular velocity and angular acceleration of a line contained within a body using the following procedure.

1. Locate point on the body using position coordinate s , which is measured from a fixed origin.
2. From a fixed reference line, measure the angular position θ of a line lying in the body. Using the dimensions of the body, relate s to θ , e.g., $s = f(\theta)$.
3. Take the first time derivative of $s = f(\theta)$ to get a relationship between v and ω .
4. Take the second time derivative to get a relationship between a and α .

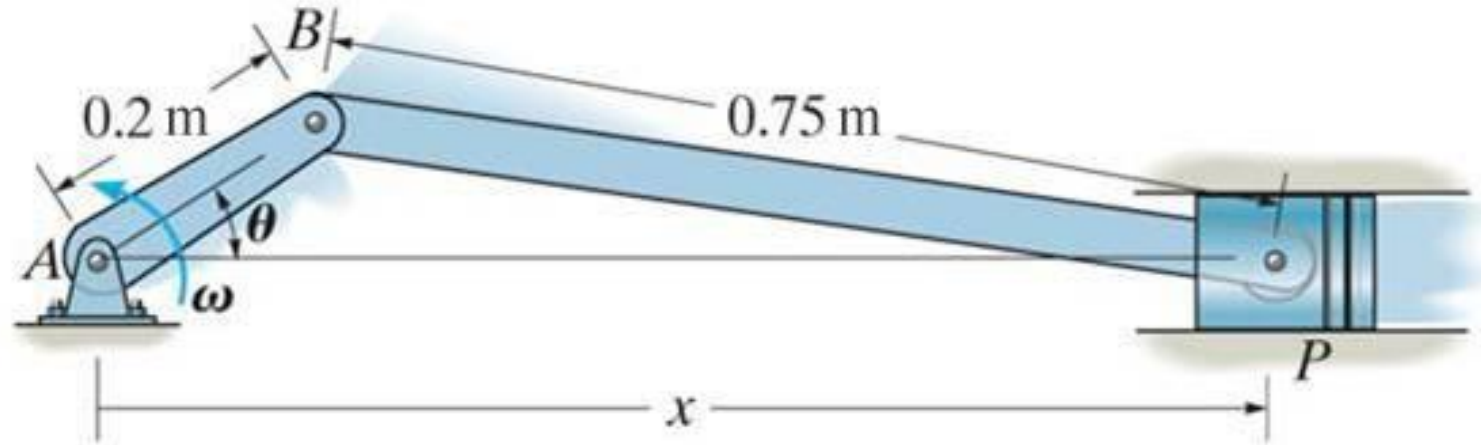
Example 1



Given: A circular cam is rotating clockwise about O with a constant ω .

Find: The velocity and acceleration of the follower rod A as a function of θ .

Example 2



Given: Crank AB rotates at a constant velocity of $\omega = 150 \text{ rad/s}$.

Find: The velocity of point P when $\theta = 30^\circ$.

Quiz

1. The position, s , is given as a function of angular position, θ , as $s = 10 \sin 2\theta$. The velocity, v , is

A) $20 \cos 2\theta$ B) $20 \sin 2\theta$

C) $20 \omega \cos 2\theta$ D) $20 \omega \sin 2\theta$

2. If $s = 10 \sin 2\theta$, the acceleration, a , is

A) $20 \alpha \sin 2\theta$

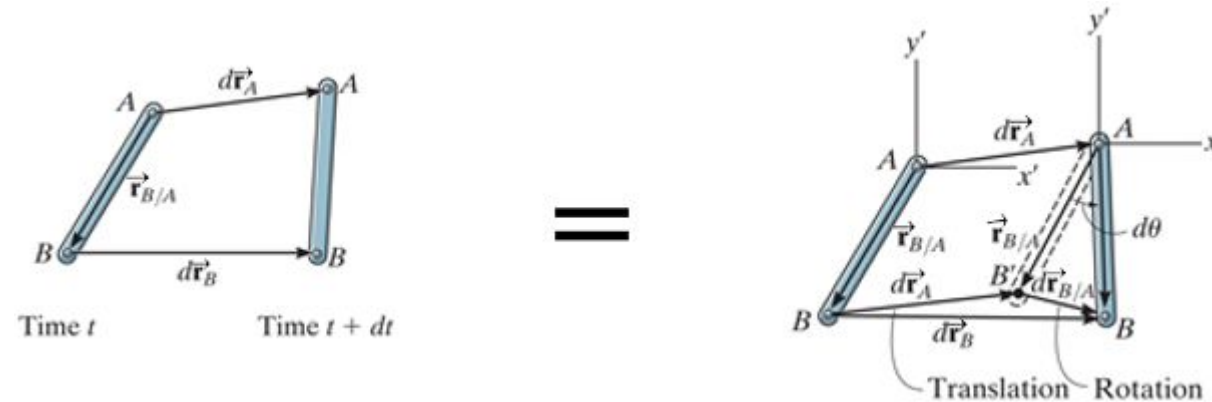
B) $20 \alpha \cos 2\theta - 40 \omega^2 \sin 2\theta$

C) $20 \alpha \cos 2\theta$

D) $-40 \alpha \sin 2\theta$

Relative motion analysis

When a body is subjected to general plane motion, it undergoes a combination of **translation** and **rotation**.

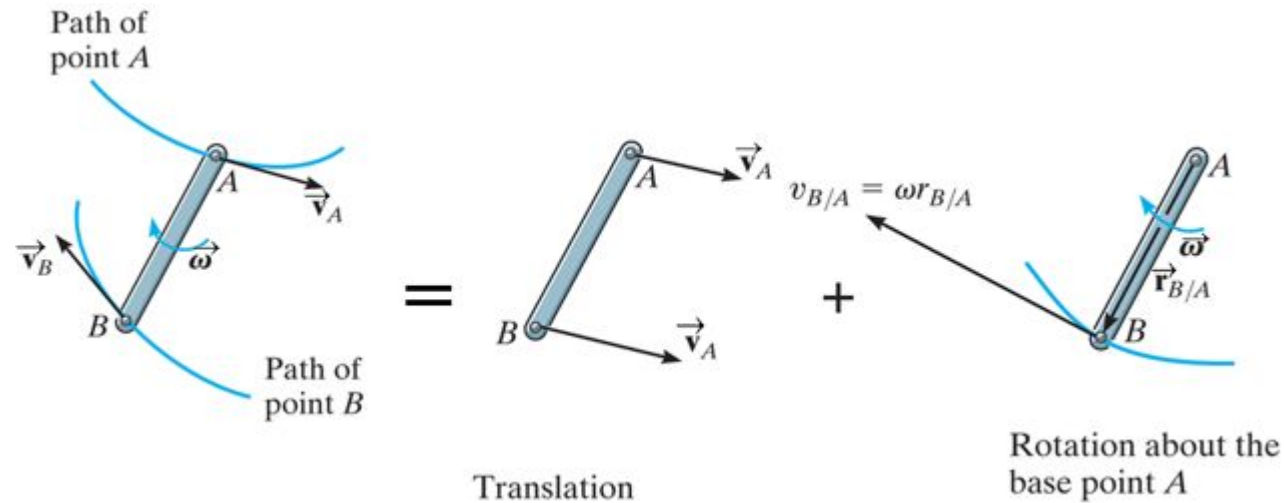


Point A is called the **base point** in this analysis. It generally has a **known** motion. The x' - y' frame translates with the body, but does not rotate. The displacement of point B can be written:

$$\text{Disp. due to translation and rotation} = \text{Disp. due to translation} + \text{Disp. due to rotation}$$

The equation is represented by vectors in circles: $d\vec{r}_B = d\vec{r}_A + d\vec{r}_{B/A}$. Arrows point from the text labels to their corresponding terms in the equation.

Relative motion analysis: velocity



The velocity at B is given as : $(d\mathbf{r}_B/dt) = (d\mathbf{r}_A/dt) + (d\mathbf{r}_{B/A}/dt)$ or

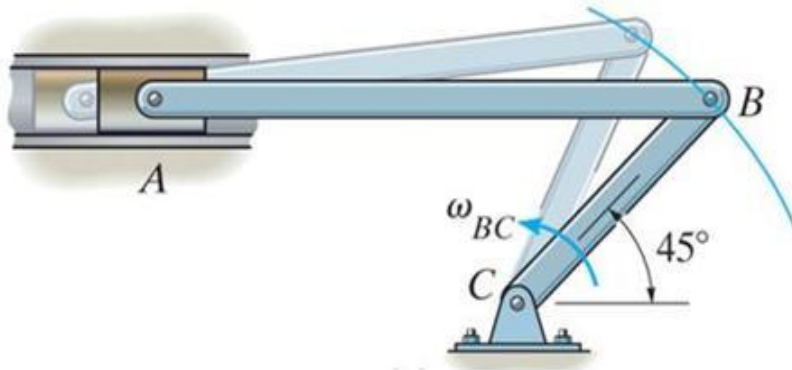
$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

Since the body is taken as rotating about A,

$$\mathbf{v}_{B/A} = d\mathbf{r}_{B/A}/dt = \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

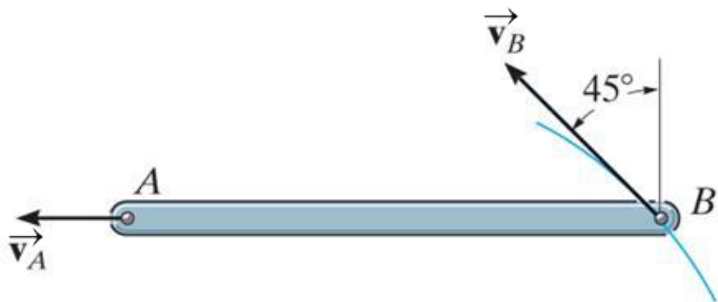
Here $\boldsymbol{\omega}$ will only have a $\hat{k}$ component since the axis of rotation is perpendicular to the plane of translation.

Relative motion analysis: velocity



$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

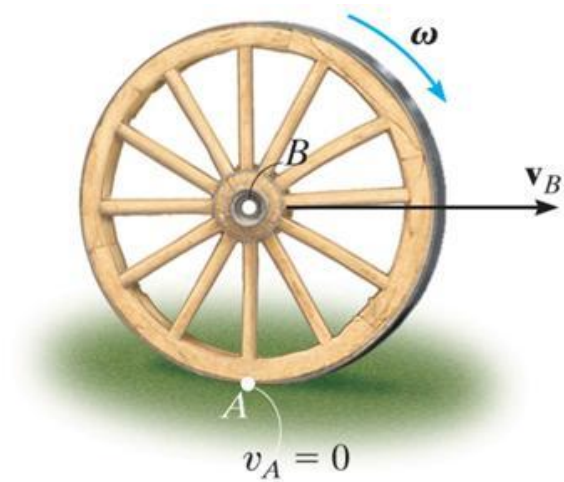
When using the relative velocity equation, points A and B should generally be points on the body with a **known motion**. Often these points are pin connections in linkages.



For example, point A on link AB must move along a horizontal path, whereas point B moves on a circular path.

The directions of $\mathbf{v}_A$ and $\mathbf{v}_B$ are known since they are always tangent to their paths of motion.

Relative motion analysis: velocity



$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

When a wheel rolls without slipping, point A is often selected to be at the point of contact with the ground.

Since there is no slipping, point A has zero velocity.

Procedure for analysis

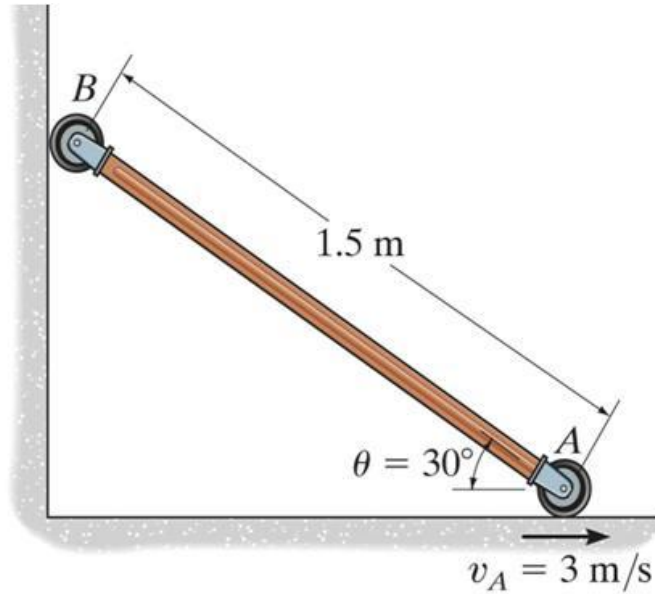
1. Establish the fixed x - y coordinate directions and draw the **kinematic diagram** of the body, showing the vectors $\mathbf{v}_A$, $\mathbf{v}_B$, $\mathbf{r}_{B/A}$ and $\boldsymbol{\omega}$. If the magnitudes are unknown, the sense of direction may be assumed.
2. Express the vectors in **Cartesian vector form** (CVN) and substitute them into $\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$. Evaluate the cross product and equate respective $\hat{i}$ and $\hat{j}$ components to obtain **two** scalar equations.
3. If the solution yields a **negative** answer, the sense of direction of the vector is **opposite** to that assumed.

Quiz

1. When a relative-motion analysis involving two sets of coordinate axes is used, the $x' - y'$ coordinate system will
 - A) be attached to the selected point for analysis.
 - B) rotate with the body.
 - C) not be allowed to translate with respect to the fixed frame.
 - D) None of the above.

2. In the relative velocity equation, $\mathbf{v}_{B/A}$ is
 - A) the relative velocity of B with respect to A.
 - B) due to the rotational motion.
 - C) $\boldsymbol{\omega} \times \mathbf{r}_{B/A}$.
 - D) All of the above.

Example 1



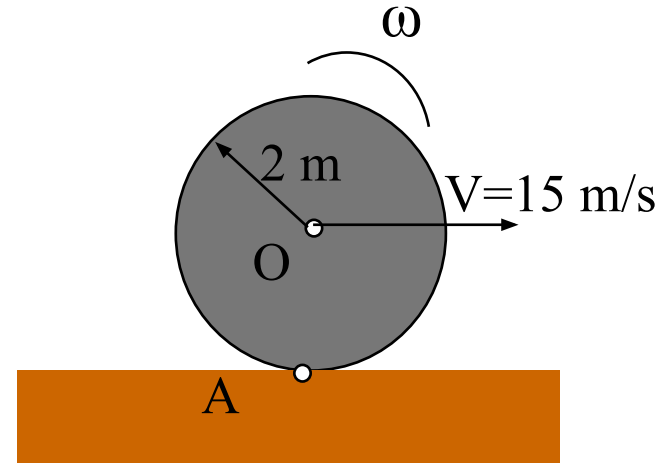
Given: Roller A is moving to the right at 3 m/s .

Find: The velocity of B at the instant $\theta = 30^\circ$.

Quiz

1. If the disk is moving with a velocity at point O of 15 m/s and $\omega = 2$ rad/s, determine the velocity at A.

- A) 0 m/s B) 4 m/s
C) 15 m/s D) 11 m/s

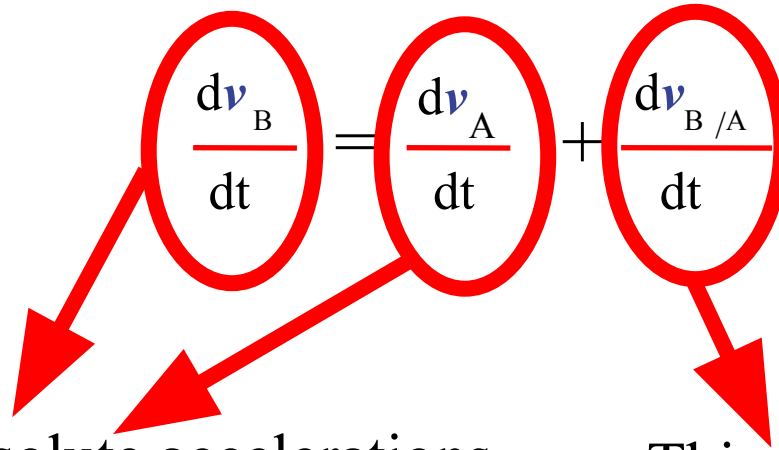


2. If the velocity at A is zero, then determine the angular velocity, ω .

- A) 30 rad/s B) 0 rad/s
C) 7.5 rad/s D) 15 rad/s

Relative motion analysis: acceleration

The equation relating the accelerations of two points on the body is determined by differentiating the velocity equation with respect to time.

$$\frac{d\mathbf{v}_B}{dt} = \frac{d\mathbf{v}_A}{dt} + \frac{d\mathbf{v}_{B/A}}{dt}$$


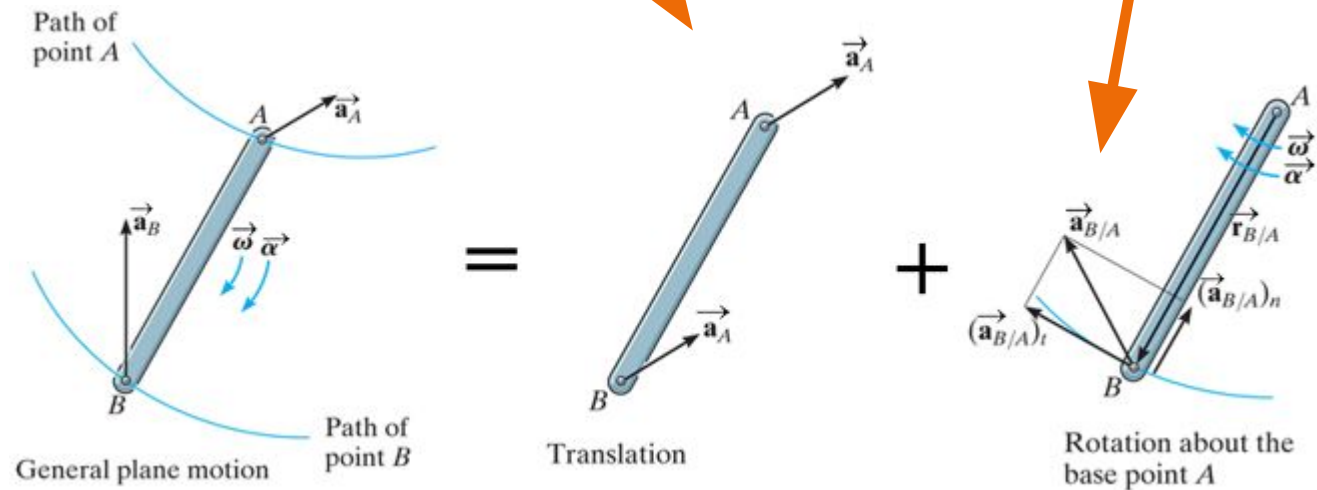
These are absolute accelerations of points A and B. They are measured from a set of fixed x,y axes.

This term is the acceleration of B with respect to A and includes both **tangential** and **normal** components.

The result is $\mathbf{a}_B = \mathbf{a}_A + (\mathbf{a}_{B/A})_t + (\mathbf{a}_{B/A})_n$

Relative motion analysis: acceleration

Graphically: $\vec{a}_B = \vec{a}_A + (\vec{a}_{B/A})_t + (\vec{a}_{B/A})_n$



The relative tangential acceleration component $(\vec{a}_{B/A})_t$ is $(\vec{\alpha} \times \vec{r}_{B/A})$ and perpendicular to $\vec{r}_{B/A}$.

The relative normal acceleration component $(\vec{a}_{B/A})_n$ is $(-\omega^2 \vec{r}_{B/A})$ and the direction is always from B towards A.

Relative motion analysis: acceleration

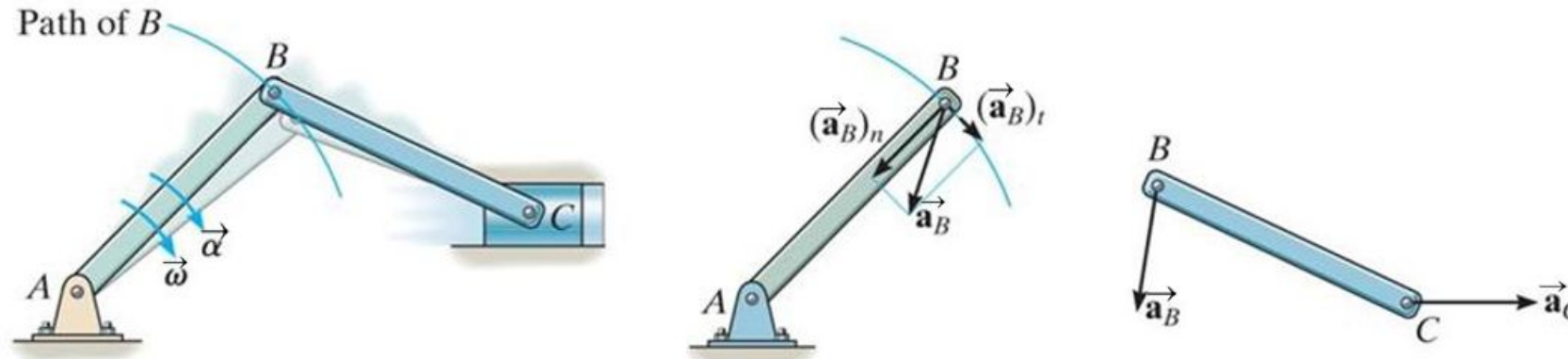
Since the relative acceleration components can be expressed as $(\mathbf{a}_{B/A})_t = \boldsymbol{\alpha} \times \mathbf{r}_{B/A}$ and $(\mathbf{a}_{B/A})_n = -\omega^2 \mathbf{r}_{B/A}$, the relative acceleration equation becomes

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A}$$

Note that the **last term** in the relative acceleration equation is **not** a cross product. It is the product of a scalar (square of the magnitude of angular velocity, ω^2) and the relative position vector, $\mathbf{r}_{B/A}$.

Application of the relative acceleration eqn

In applying the relative acceleration equation, the two points used in the analysis (A and B) should generally be selected as points which have a **known motion**, such as **pin connections** with other bodies.



In this mechanism, point B is known to travel along a **circular path**, so $\vec{a}_B$ can be expressed in terms of its normal and tangential components. Note that point B on link BC will have the **same acceleration** as point B on link AB.

Point C, connecting link BC and the piston, moves along a **straight-line path**. Hence, $\vec{a}_C$ is directed horizontally.

Procedure for analysis

1. Establish a fixed coordinate system.
2. Draw the kinematic diagram of the body.
3. Indicate on it $\mathbf{a}_A$, $\mathbf{a}_B$, $\boldsymbol{\omega}$, $\boldsymbol{\alpha}$, and $\mathbf{r}_{B/A}$. If the points A and B move along curved paths, then their accelerations should be indicated in terms of their tangential and normal components, i.e., $\mathbf{a}_A = (\mathbf{a}_A)_t + (\mathbf{a}_A)_n$ and $\mathbf{a}_B = (\mathbf{a}_B)_t + (\mathbf{a}_B)_n$.
4. Apply the relative acceleration equation:

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A}$$

5. If the solution yields a negative answer for an unknown magnitude, this indicates that the sense of direction of the vector is opposite to that shown on the diagram.

Quiz

1. If two bodies contact one another without slipping, and the points in contact move along different paths, the tangential components of acceleration will be _____ and the normal components of acceleration will be _____.
A) the same, the same B) the same, different
C) different, the same D) different, different

2. When considering a point on a rigid body in general plane motion,
A) Its total acceleration consists of both absolute acceleration and relative acceleration components.
B) It's total acceleration consists of only absolute acceleration components.
C) Its relative acceleration component is always normal to the path.
D) None of the above.

Summary

Rigid body

Planar motion of rigid body (curvilinear translation + rotation)

Angular distance, velocity and acceleration

Rotational motion analysis

Absolute motion analysis